

1. a, d is proposition, (a) is true
d is false

2. (a) (b) (c) (d) (e)
F T T F T

3. (a)
(a) $r \wedge \neg q$
(b) $p \wedge q \wedge r$
(c) $r \rightarrow p$
(d) $p \wedge \neg q \wedge r$
(e) $(p \wedge q) \rightarrow r$

(a) $\neg p$
(b) $p \wedge \neg q$
(c) $p \rightarrow q$
(d) $p \rightarrow q$
(e) $q \rightarrow p$

4. (a)

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

(b)

p	q	$\neg p$	$p \rightarrow q$	$\neg p \rightarrow (p \rightarrow q)$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

(c)

p	q	$p \wedge q$	$p \rightarrow q$	$(p \wedge q) \rightarrow (p \rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

f. d)

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	F	F	T

(e)

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

$(p \rightarrow q) \wedge (q \rightarrow r)$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
T	T
F	T
F	T
F	T
F	T
F	T
F	T
F	T

5. a)

$$\begin{aligned}
 (p \wedge q) \rightarrow p &\equiv \neg(p \wedge q) \vee p \\
 &\equiv (\neg p \vee \neg q) \vee p \\
 &\equiv \neg q \vee (\neg p \vee p) \\
 &\equiv \neg q \vee \top \\
 &\equiv \top
 \end{aligned}$$

b)

$$\begin{aligned}
 \neg p \rightarrow (p \rightarrow q) &\equiv p \vee (p \rightarrow q) \\
 &\equiv p \vee (\neg p \vee q) \\
 &\equiv (p \vee \neg p) \vee q \\
 &\equiv \top \vee q \\
 &\equiv \top
 \end{aligned}$$

c)

$$\begin{aligned}
 (p \wedge q) \rightarrow (p \rightarrow q) &\equiv \neg(p \wedge q) \vee (p \rightarrow q) \\
 &\equiv \neg(p \wedge q) \vee (\neg p \vee q) \\
 &\equiv (\neg p \vee \neg q) \vee (\neg p \vee q) \\
 &\equiv (\neg p \vee \neg p) \vee (\neg q \vee q) \\
 &\equiv (\neg p \vee \neg p) \vee \top \\
 &\equiv \top
 \end{aligned}$$

d)

$$\begin{aligned}
 [\neg p \wedge (p \vee q)] \rightarrow q &\equiv \neg[\neg p \wedge (p \vee q)] \vee q \\
 &\equiv p \vee \neg(p \vee q) \vee q \\
 &\equiv (p \vee q) \vee \neg(p \vee q) \\
 &\equiv \top
 \end{aligned}$$

e)

$$\begin{aligned}
 [(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r) &\equiv \neg[(p \rightarrow q) \wedge (q \rightarrow r)] \vee (p \rightarrow r) \\
 &\equiv \neg[(\neg p \vee q) \wedge (\neg q \vee r)] \vee (\neg p \vee r) \\
 &\equiv [(\neg p \vee q) \vee \neg(\neg q \vee r)] \vee (\neg p \vee r) \\
 &\equiv [(\neg p \vee q) \vee (q \vee \neg r)] \vee (r \vee \neg p) \\
 &\equiv (p \vee \neg r) \vee (r \vee \neg p) \\
 &\equiv (p \vee \neg r) \vee \neg(p \vee \neg r) \equiv \top
 \end{aligned}$$

6. (a) False

(b) True

(c) False

(v)

(a)

$$\exists x [(C(x) \wedge F(x)) \wedge \neg D(x)]$$

(b)

$$\forall x \neg [C(x) \vee (F(x) \vee D(x))]$$

7.

(1)

$$\forall x [(P(x) \vee Q(x)) \wedge (\neg Q(x) \vee S(x)) \wedge (R(x) \rightarrow S(x))]$$

$$\therefore \exists x \neg P(x)$$

$$\equiv (\neg P(x) \rightarrow Q(x)) \wedge (Q(x) \rightarrow S(x)) \wedge (S(x) \rightarrow \neg R(x))$$

$$\therefore \exists x \neg R(x)$$

$$\equiv \neg P(x) \rightarrow \neg R(x)$$

(2)

Let "Allen is a good boy" is p

"David is happy" is q

"Hillary is a good girl" is r

$$\Rightarrow (\neg p \vee r) \wedge (p \vee q) \rightarrow (r \vee q) \text{ By Resolution}$$

QED.

8.

n is an integer and $\frac{n^3+5 \text{ is odd}}{P}$, then $\frac{n \text{ is even}}{Q}$

(a)

By contraposition: $(\neg Q \rightarrow \neg P) \rightarrow (P \rightarrow Q)$

Assume $\frac{n \text{ is odd}}{\neg Q}$, $\frac{n^3+5 \text{ is even}}{\neg P}$

$$n = 2k+1, k \in \mathbb{Z}$$

$$n^3+5 = (2k+1)^3+5 = 8k^3+1+12k^2+6k+5 = 8k^3+12k^2+6k+6 = 2(4k^3+6k^2+3k+3),$$

$$= 2j, j = 4k^3+6k^2+3k+3$$

$$\Rightarrow n^3+5 \text{ is even}$$

$$\therefore \neg Q \rightarrow \neg P$$

$$\therefore P \rightarrow Q$$

QED.

(b)

By contradiction: $P \wedge \neg Q$ is false, then $P \rightarrow Q$

Assume $\frac{n^3+5 \text{ is odd}}{P}, \frac{n \text{ is odd}}{Q}$

$$n = 2k+1, k \in \mathbb{Z}$$

$$n^3+5 = 8k^3+1+12k^2+6k+5 = 8k^3+12k^2+6k+6 = 2j, j = 4k^3+6k^2+3k+3$$

$\Rightarrow n^3+5$ is even, not odd.

$\Rightarrow \neg P$.

$\therefore P \wedge \neg Q$ is false

$\therefore P \rightarrow Q$ Q.E.D.

9.

Proof $A \subseteq B \Leftrightarrow \bar{B} \subseteq \bar{A}$

$$A \subseteq B \equiv \forall x [(x \in A) \rightarrow (x \in B)] \quad \text{By UI}$$

$$\equiv \forall x [\neg(x \in B) \rightarrow \neg(x \in A)] \quad \text{By contraposition}$$

$$\equiv \forall x [(x \in \bar{B}) \rightarrow (x \in \bar{A})] \quad \text{By definition.}$$

$$\equiv \bar{B} \subseteq \bar{A} \quad \text{By UG}$$

$$\Rightarrow A \subseteq B \Leftrightarrow \bar{B} \subseteq \bar{A}$$

10.

$$(A-B) \cap (B-C) \cap (A-C) \equiv (A \cap \bar{B}) \cap (B \cap \bar{C}) \cap (A \cap \bar{C})$$

$$\equiv (A \cap A) \cap (\bar{B} \cap B) \cap (\bar{C} \cap \bar{C})$$

$$\equiv A \cap \phi \cap \bar{C}$$

$$\equiv \phi \quad \text{Q.E.D.}$$

11. (i) $| - | : f(a) = f(b) \Leftrightarrow a = b$

(a)

$$f(a) = f(b)$$

$$a^2+1 = b^2+1$$

$$a^2 - b^2 = 0$$

$$a^2 = b^2$$

$$a = \pm b$$

$$\Rightarrow f(n) = n^2 + 1$$

is NOT 1-1 function.

(b)

$$f(a) = f(b)$$

$$a^3 = b^3$$

$$a = b$$

$$\Rightarrow f(n) = n^3$$

is 1-1 function.

11. e)

a)

$$f(m, n) = m^2 - n^2 = (m+n)(m-n)$$

eg. $f(m, n) \neq 2$. $\Rightarrow f$ is NOT onto

b).

$$f(m, n) = m + n$$

$\forall x \in \mathbb{Z}, \exists (2x, -x) \in \mathbb{Z} \times \mathbb{Z}$ s.t.

$$f(2x, -x) = x$$

$\Rightarrow f$ is onto

c).

$$f(m, n) \neq 4$$

$f(m, n) \geq 0 \Rightarrow f$ is not onto